



CH101

Class 07; Physical Chemistry

Hydrogenic Atoms

Principal quantum number: n ; takes the values of 1, 2, 3, ...

‘ n ’ decides the values of energy

$$E_n = -\frac{Z^2 \mu e^4}{32\pi^2 \epsilon_0^2 \hbar^2 n^2}$$

The angular solutions provide the other two quantum numbers

‘ l ’ is the azimuthal quantum number or orbital angular momentum quantum number

An electron in an orbital with quantum number m_l has the angular momentum of magnitude $\sqrt{l(l+1)}\hbar$ with $l = 0, 1, 2, \dots, n-1$. *Note how the principal quantum number n decides the values of l .*

An electron in an orbital with quantum number m_l has a z-component of angular momentum $m_l \hbar$; with $m_l = 0, \pm 1, \pm 2, \pm 3, \dots, \pm l$. *Note how the quantum number l decides the values of m_l .*

In order to define a state of electron in hydrogenic atoms one needs to consider also the spin of electron. Electron possesses an intrinsic angular momentum that is described by two quantum numbers ‘ s ’ and ‘ m_s ’.

The value of ‘ s ’ is fixed at $\frac{1}{2}$ for an electron.

‘ m_s ’ can be either $+\frac{1}{2}$ or $-\frac{1}{2}$.

The state of an electron in hydrogenic atoms is defined by n, l, m_l and m_s .

The energy levels of an electron in hydrogenic atoms are defined by

$$E_n = -\frac{Z^2 \mu e^4}{32\pi^2 \epsilon_0^2 \hbar^2 n^2} \text{ Thus } E_n \propto Z^2 \text{ and hence the ground state of He}^+ \text{ is four times}$$

deeper than that in H.

The energy states here are referred to as the bound states of the atoms in which the energy of the electron is lower in any of these states than if the electron and nucleus were infinitely separated.

From the spectroscopic results of hydrogen atom we have the Rydberg constant

$$E_{n_1} - E_{n_2} = R_H \left(\frac{1}{n_2^2} - \frac{1}{n_1^2} \right) = hcR_H = \frac{\mu_H e^4}{32\pi^2 \epsilon_0^2 \hbar^2}$$

$$\text{Thus } R_H = \frac{\mu_H e^4}{8 \epsilon_0^2 \hbar^3 c} \text{ and In general, } R = \frac{m_e e^4}{8 \epsilon_0^2 \hbar^3 c}$$

$$\text{Hence } R_H = \frac{\mu_H}{m_e} R$$

Do it now:

Calculate R_H from the following fundamental constants

Mass of electron, $m_e = 9.109 \, 38 \times 10^{-31} \text{ kg}$

Charge of electron, $e = 1.602 \, 176 \times 10^{-19} \text{ C}$

$\epsilon_0 = 8.854 \, 19 \times 10^{-12} \text{ J}^{-1} \text{ C}^2 \text{ m}^{-1}$

$\hbar = 6.626 \, 08 \times 10^{-34} \text{ J s}$

$c = 2.997 \, 924 \, 58 \times 10^8 \text{ m s}^{-1}$

Answer: $1.097 \, 37 \times 10^5 \text{ cm}^{-1}$.

Atomic Orbitals:

Consider the ground state orbital of the hydrogenic atom

$$n = 1, l = 0, m_l = 0$$

$$R = 2 \left(\frac{Z}{a_0} \right)^{3/2} e^{-\rho/2}; \text{ where } \rho = \frac{2Zr}{a_0} \text{ and } Y_{l,m_l}(\theta, \phi) = \sqrt{\frac{1}{4\pi}}$$

The wavefunction, for $Z = 1$, can then be written as:

$$\Psi = \frac{1}{(\pi a_0^3)^{1/2}} e^{-r/a_0}; \text{ The wavefunction is independent of angle and has the same}$$

for all angles at a particular value of r .

Radial distribution function:

$|\Psi|^2$ is the probability density of finding an electron in any region.

For the electron in the ground state $|\Psi|^2 \propto e^{-2Zr/a_0}$

Now consider the probability of finding the electron anywhere on a spherical shell with thickness dr and radius r .

The volume of the shell is $4\pi r^2 dr$

The probability that the electron will be found anywhere between the distance r and $r+dr$ and in the shell is $|\Psi|^2 4\pi r^2 dr = P(r)dr$; where $P(r) = 4\pi r^2 |\Psi|^2$

The above expression is Radial Probability and is valid for electrons in orbitals that are spherically symmetric.

When the orbital is **not** spherically symmetric then,

$$\begin{aligned} P(r)dr &= \int_0^\pi \int_0^{2\pi} R(r)^2 |Y(\theta, \phi)|^2 r^2 dr \sin \theta d\theta d\phi \\ &= r^2 R(r)^2 dr \int_0^\pi \int_0^{2\pi} |Y(\theta, \phi)|^2 \sin \theta d\theta d\phi = r^2 R(r)^2 dr \end{aligned}$$

The last equation follows that the spherical harmonics are normalized.

Hence the expression for probability density of non-spherically symmetric wavefunctions is

$$P(r) = r^2 R(r)^2$$

For a 1s orbital of hydrogenic atoms,

$$P(r) = \frac{4Z^3}{a_0^3} r^2 e^{-2Zr/a_0}$$

The wavefunction for a 2p orbital with $m_l=0$ is

$$\begin{aligned}\Psi_{p_0} &= R_{2,1}(r)Y_{1,0}(\theta, \varphi) = \frac{1}{4(2\pi)^{1/2}} \left(\frac{Z}{a_0}\right)^{5/2} r \cos \theta e^{-Zr/2a_0} \\ &= r \cos \theta f(r) \\ \Psi_{p_z} &= z f(r)\end{aligned}$$

This orbital is called 2p_z orbital and has a node in the x-y plane

The wavefunctions of 2p orbitals with $m_l=\pm 1$ have the following form.

$$\Psi_{p_{\pm 1}} = R_{2,1}(r)Y_{1,\pm 1}(\theta, \varphi) = \mp \frac{1}{8(\pi)^{1/2}} \left(\frac{Z}{a_0}\right)^{5/2} r e^{-Zr/2a_0} \sin \theta e^{\pm i\phi}$$

The functions above correspond to a ϕ -dependent function for a particle with angular momentum either clockwise or anti-clockwise around the Z-axis. They have zero amplitude when $\theta=0$ and 180° and maximum amplitude at 90° which is in xy-plane. The linear combination of the above functions would generate real functions.

$$\begin{aligned}\Psi_{p_x} &= -\frac{1}{2^{1/2}}(p_{+1} + p_{-1}) = r \sin \theta \cos \phi f(r) = x f(r) \\ \Psi_{p_y} &= \frac{i}{2^{1/2}}(p_{+1} - p_{-1}) = r \sin \theta \sin \phi f(r) = y f(r)\end{aligned}$$

These linear combinations are standing waves with no net orbital angular momentum around the z-axis, as they are superposition of states with equal and opposite values of angular m_l . The p_x orbital has the same shape as the p_z orbital but directed along the x-axis. Similarly, the p_y orbital has the same shape as the p_z orbital but directed along the y-axis.

d-orbitals

When $n = 3$, l can be 0, 1 or 2. The shell consists of one 3s orbital, three 2p orbitals and five 3d orbitals. The five d orbitals have $m_l = +2, +1, 0, -1, -2$ and correspond to five different angular momenta around the z-axis (but the magnitude of the momentum is the same as $l=2$).

$$d_{xy} = xyf(r); \quad d_{yz} = yzf(r) \quad d_{zx} = zxf(r)$$

$$d_{x^2-y^2} = \frac{1}{2}(x^2 - y^2)f(r)$$

$$d_{z^2} = \frac{1}{2\sqrt{3}}(3z^2 - r^2)f(r)$$